

Hypothesis Testing in Bioinformatics

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Introduction to Bioinformatics

Presentation Overview

- 1 Fundamentals of Statistical Testing
- 2 Specific Hypothesis Tests
- 3 Multiple Testing Correction

What is Statistics?

- **Statistics** is the science of **collecting, analyzing, interpreting, and presenting data**.
- It provides the tools to draw reliable conclusions from data, especially when dealing with **uncertainty** and **variability**.
- **In Bioinformatics:** Statistics is essential for:
 - Identifying **Differentially Expressed Genes** (DEGs).
 - Predicting protein structure or function with an associated **confidence score**.
 - Determining the **significance** of sequence alignments (e.g., BLAST E-value).

The Concept of a Statistic

- A **Statistic** (or **Test Statistic**) is a numerical summary of a **sample**.
- It is used to estimate a characteristic of the entire **population**, known as a **parameter**.

Examples and Formulas:

- **Sample Mean** ($\bar{x}$): Used to estimate the population mean (μ).

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

- **Test Statistic** (Z , t , etc.): A measure of how far a sample result is from the null hypothesis, often in terms of standard error.

$$\text{Test Statistic} = \frac{\text{Observed Sample Statistic} - \text{Hypothesized Parameter}}{\text{Standard Error of the Statistic}}$$

Null and Alternative Hypothesis (H_0 and H_1)

Hypothesis testing requires two opposing statements about a population parameter:

- **Null Hypothesis (H_0):**

- Assumes the **status quo** or **no effect/difference**. It is the hypothesis we test directly.
- Always includes an equality sign ($\mu_1 = \mu_2, p \leq 0.5$).
- **Bioinformatics Example:** The mean expression of Gene A is the same in cancer and normal tissues ($\mu_{\text{cancer}} = \mu_{\text{normal}}$).

- **Alternative Hypothesis (H_1 or H_a):**

- Represents the effect or difference the researcher believes is true.
- Uses inequality signs ($\neq, <, \text{ or } >$).
- **Bioinformatics Example:** The mean expression of Gene A is higher in cancer tissues ($\mu_{\text{cancer}} > \mu_{\text{normal}}$).

P-value (Probability Value)

Definition:

- The **P-value** is the probability of observing test results at least as extreme as the results actually observed, **assuming the null hypothesis (H_0) is true**.

Decision Rule (based on the Significance Level α):

- If $P\text{-value} \leq \alpha$ (e.g., 0.05), we conclude the result is **statistically significant**. **Reject H_0** .
- If $P\text{-value} > \alpha$, we **Fail to Reject H_0** .

Interpretation Example:

- If a test on Gene X yields a P-value of 0.001, it means: *There is only a 0.1% chance of seeing this difference in expression if the gene was truly not differentially expressed (H_0 is true).*

1. Permutation Test (Non-parametric)

Concept:

- A non-parametric test that **does not assume** a specific distribution (like Normal) for the population.
- It's an **exact test** when all permutations are calculated, or an approximation when using a large random sample of permutations.
- **Core Principle:** If the null hypothesis (e.g., $\mu_1 = \mu_2$) is true, the group labels are arbitrary, and the data could have occurred under any permutation of the labels.

1. Permutation Test (Non-parametric) - Cont.

Procedure:

- 1 Calculate the observed test statistic (T_{obs}) from the original data.
- 2 Pool the data from all groups.
- 3 **Permute** (shuffle) the combined data and randomly reassign the group labels.
- 4 Calculate the test statistic (T^*) for the permuted data. Repeat thousands of times.
- 5 The P-value is the proportion of T^* values that are as extreme as or more extreme than T_{obs} .

Permutation Test Formula

The approximate P-value is calculated by counting how often the shuffled statistic is in the critical region defined by the observed statistic:

$$\text{P-value} \approx \frac{\text{Number of permutations with } |T^*| \geq |T_{\text{obs}}|}{\text{Total number of permutations } (N)}$$

Example (Bioinformatics):

- **Data:** Gene expression levels for 5 patients (Group A) and 5 controls (Group B).
- T_{obs} : Difference in mean expression ($\bar{x}_A - \bar{x}_B$).
- **Permutations:** There are $\binom{10}{5} = 252$ ways to split the 10 data points into two groups of 5.
- **Conclusion:** If 5 out of 252 permutations yield a difference larger than T_{obs} , the P-value is $5/252 \approx 0.0198$. If $\alpha = 0.05$, we reject H_0 .

2. Z-test Introduction

A Z-test is a type of hypothesis test that compares the sample's average to the population's average and calculates the Z-score and tells us how much the sample average is different from the population average by looking at how much the data normally varies. It is particularly useful when the sample size is large. This Z-Score is also known as Z-Statistics formula is:

$$\text{Z-Score} = \frac{\bar{X} - \mu}{\sigma}$$

where,

$\bar{X}$: mean of the sample.

μ : mean of the population.

σ : Standard deviation of the population.

Z-Score Example

The average family annual income in India is 200k with a standard deviation of 5k and the average family annual income in Delhi is 300k. Then Z-Score for Delhi will be.

$$\text{Z-Score} = \frac{300 - 200}{5} = 20$$

This indicates that the average family's annual income in Delhi is 20 standard deviations above the mean of the population (India).

Assumptions of Z-test

For a z-test to provide reliable results these assumptions must be met:

- Normal Distribution: The population from which the sample is drawn should be approximately normally distributed.
- Equal Variance: The samples being compared should have the same variance.
- Independence: All data points should be independent of one another.

Steps to Perform Z-test

1. First we identify the null and alternate hypotheses.
2. Then we determine the level of significance (α).
3. Next we find the critical value of Z in the z-test.
4. Then we calculate the z-test statistics using the formula :

$$Z = \frac{(\bar{X} - \mu)}{(\sigma / \sqrt{n})}$$

Where:

$\bar{X}$: mean of the sample.

μ : mean of the population.

σ : Standard deviation of the population.

n : sample size. 5. Now we compare with the hypothesis and decide whether to reject or not reject the null hypothesis.

Types of Z-test

There are mainly two types of Z-tests.

1. One Sample Z test
2. Two-sampled z-test

One Sample Z-test

A one-sample Z-test is used to determine if the mean of a single sample is significantly different from a known population mean.

When to Use:

- The population standard deviation is known.
- The sample size is large (usually $n > 30$).
- The data is approximately normally distributed.

One-Sample Z-test Example

Suppose a company claims that their new smartphone has an average battery life of 12 hours. A consumer group tests 100 phones and finds an average battery life of 11.8 hours with a known population standard deviation of 0.5 hours.

One-Sample Z-test Example Solution

Step 1: Hypotheses:

$$H_0 : \mu = 12$$

$$H_1 : \mu \neq 12$$

Step 2: Calculate the Z-Score:

$$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

Where: $\bar{x} = 11.8$, $\mu = 12$, $\sigma = 0.5$, $n = 100$

After substituting the values we get:

$$z = \frac{11.8 - 12}{\frac{0.5}{\sqrt{100}}} = -4$$

One-Sample Z-test Decision

Step 3: Decision Since $|Z| = 4 > 1.96$ (critical value for $\alpha = 0.05$), so we reject H_0 indicate significant evidence against the company's claim.

Two-Sampled Z-test

In this test we have provided 2 normally distributed and independent populations and we have drawn samples at random from both populations. Here we consider μ_1 and μ_2 to be the population mean and X_1 and X_2 to be the observed sample mean. Here our null hypothesis could be like this:

$$H_0 : \mu_1 - \mu_2 = 0$$

and alternative hypothesis

$$H_1 : \mu_1 - \mu_2 \neq 0$$

Two-Sample Z-test Formula

The formula for calculating the z-test score:

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

where σ_1 and σ_2 are the standard deviation and n_1 and n_2 are the sample size of population corresponding to μ_1 and μ_2 .

Two-Sample Z-test Example

There are two groups of students preparing for a competition: Group A and Group B. Group A has studied offline classes, while Group B has studied online classes. After the examination the score of each student comes. Now we want to determine whether the online or offline classes are better. **Group A:** Sample size = 50, Sample mean = 75, Sample standard deviation = 10

Group B: Sample size = 60, Sample mean = 80, Sample standard deviation = 12

Assuming a 5% significance level, perform a two-sample z-test to determine if there is a significant difference between the online and offline classes.

Two-Sample Z-score Example Solution

Step 1: Null & Alternate Hypothesis

Null Hypothesis:

$$\mu_1 - \mu_2 = 0$$

Alternate Hypothesis:

$$\mu_1 - \mu_2 \neq 0$$

Step 2: Significance Level $\alpha = 0.05$

Two-Sample Z-score Example Solution

Step 3: Z-Score

$$Z = \frac{(75 - 80) - 0}{\sqrt{\frac{10^2}{50} + \frac{12^2}{60}}} = \frac{-5}{2.0976} \approx -2.384$$

Two-Sample Z-test Example Solution

Step 4: Check Critical Z-Score value in the Z-Table for $\frac{\alpha}{2} = 0.025$

Critical Z-Score = 1.96

Step 5: Compare with the absolute Z-Score value

$\text{absolute}(\text{Z-Score}) > \text{Critical Z-Score}$

So we reject the null hypothesis and there is a significant difference between the online and offline classes.

3. T-test Introduction

The t-test is used to compare the averages of two groups to see if they are significantly different from each other. Suppose you want to compare the test scores of two groups of students:

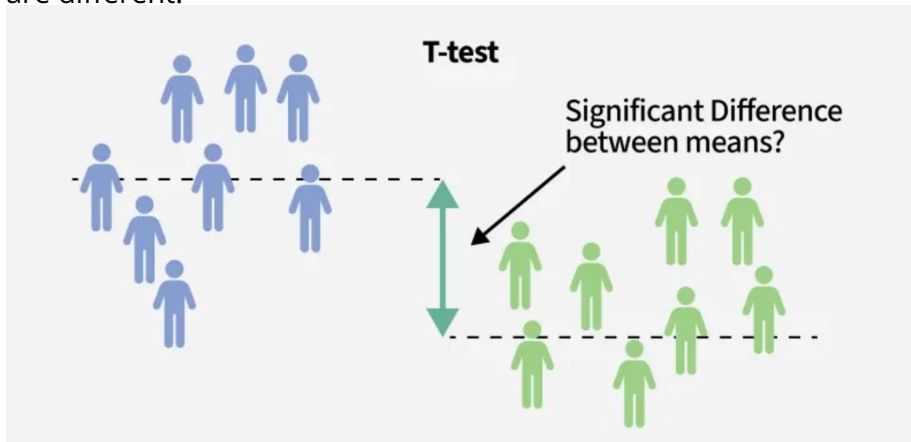
Group 1: 30 students who studied with Method A.

Group 2: 30 students who studied with Method B.

We use a t-test to check if there is a significant difference in the average test scores between the two.

T-test Introduction

The t-test is part of Hypothesis testing where we start with an assumption the null hypothesis that the two group means are the same. Then the test helps you decide if there's enough evidence to reject that assumption and conclude that the groups are different.



Assumptions in T-test

- Independence
- Normality
- Homogeneity of Variances
- Absence of Outliers

Types of T-tests

There are three types of t-tests and they are categorized as dependent and independent t-tests.

Types of t test

One sample t-test



Is there a difference between a group and the population.

Independent samples t-test



Is there a difference between two groups

Paired samples t-test



Is there a difference in a group between two points in time

One Sample T-test

One sample t-test is used for comparison of the sample mean of the data to a particularly given value. We can use this when the sample size is small (under 30), data is collected randomly and it is approximately normally distributed. It can be calculated as:

The diagram shows the formula for a one-sample t-test with green arrows pointing from descriptive labels to the corresponding parts of the formula:

$$t = \frac{\overline{X} - \mu}{\frac{S}{\sqrt{n}}}$$

- Mean of the sample**: Points to $\overline{X}$
- Reference Value**: Points to μ
- Standard deviation**: Points to S
- Number of cases**: Points to n

One Sample T-test Example

Suppose you have a random sample of 31 energy bars. The label claims that each bar contains 20 grams of protein. You measure the protein content and you want to test whether the true mean protein content is different from 20 g. Here are the (summary) statistics:

$$n = 31, \quad \bar{x} = 21.40, \quad s = 2.54$$

One Sample T-test Example Solution

Hypotheses.

$$H_0 : \mu = 20 \quad \text{vs.} \quad H_a : \mu \neq 20$$

Significance level.

Use $\alpha = 0.05$.

Test statistic. First compute the standard error (SE) of the mean:

$$SE = \frac{s}{\sqrt{n}} = \frac{2.54}{\sqrt{31}} \approx 0.456$$

Then compute the t-value:

$$t = \frac{\bar{x} - \mu_0}{SE} = \frac{21.40 - 20}{0.456} \approx 3.07$$

Degrees of freedom (df).

$$df = n - 1 = 31 - 1 = 30$$

One Sample T-test Example Solution

Critical value. For a two-tailed test at $\alpha = 0.05$ with 30 degrees of freedom, the critical t-value is approximately ± 2.043 . **Decision.** Since $t = 3.07$ is greater than $+2.043$, we reject the null hypothesis. **Conclusion.** There is sufficient evidence at the 5% significance level to conclude that the mean protein content of the energy bars is not 20 g. In fact, the sample suggests it is higher than 20 g.

4. Independent Sample T-test (Two-Sample T-test)

An Independent sample t-test commonly known as an unpaired sample t-test is used to find out if the differences found between two groups is actually significant or just a random occurrence. We can use this when:

- The population mean or standard deviation is unknown.
- The two samples are separate/independent.

The diagram shows the formula for the Independent Sample T-test with green arrows pointing to specific parts of the equation:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$

Annotations:

- Mean of the sample 1 (points to $\bar{X}_1$)
- Mean of the sample 2 (points to $\bar{X}_2$)
- Standard deviation sample 1 and 2 (points to S_1^2 and S_2^2)
- Number of cases sample 1 and 2 (points to n_1 and n_2)

Independent Sample T-test Example

Suppose a researcher wants to compare the average body mass of two species of lizards: house geckos and Carolina anoles. A sample of 10 geckos and 10 anoles are measured. The summary statistics are:

$$n_1 = 10, \quad \bar{X}_1 = 15.3, \quad s_1 = 2.2 \quad (\text{geckos})$$

$$n_2 = 10, \quad \bar{X}_2 = 12.8, \quad s_2 = 1.9 \quad (\text{anoles})$$

Independent Sample T-test Example Solution

Hypotheses.

$$H_0 : \mu_1 = \mu_2 \quad \text{vs.} \quad H_a : \mu_1 \neq \mu_2$$

Assumptions.

- The two samples are independent.
- The population distributions are approximately normal.
- Variances of the two populations are equal (or at least similar).

Test statistic. We use the formula

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Plug in the numbers:

$$t = \frac{15.3 - 12.8}{\sqrt{\frac{2.2^2}{10} + \frac{1.9^2}{10}}} = \frac{2.5}{\sqrt{0.845}} \approx 2.72$$

Independent Sample T-test Example Solution

Degrees of freedom. If we assume equal variances, the (pooled) degrees of freedom is:

$$\nu = n_1 + n_2 - 2 = 10 + 10 - 2 = 18$$

Critical value / Decision. We choose a two-tailed test at significance level $\alpha = 0.05$. From a t -distribution table or using software, $t_{18,0.025} \approx 2.101$. Because $|t| = 2.72 > 2.101$, we reject the null hypothesis. **Conclusion.** There is significant evidence (at the 5% level) to conclude that the mean body mass of house geckos and Carolina anoles differ.

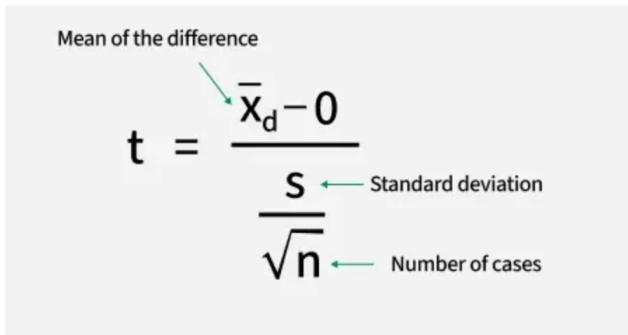
Paired Two-sample T-test (Dependent Samples)

Paired sample t-test is used to find out if the difference in the mean of two samples is 0. The test is done on dependent samples. We can use this when:

- Two similar samples are given (e.g., before/after, twins).
- The dependent variable data is continuous.
- The observations are independent of one another (within the pair).
- The dependent variable is approximately normally distributed.

Paired Two-sample T-test Formula

It can be calculated using the below equation.



The diagram shows the formula for a Paired Two-sample T-test with green arrows pointing to specific parts of the equation:

$$t = \frac{\bar{x}_d - 0}{\frac{s}{\sqrt{n}}}$$

Annotations:

- Mean of the difference (points to $\bar{x}_d$)
- Standard deviation (points to s)
- Number of cases (points to n)

Paired Two-sample T-test Example

A psychologist measures the “stress score” of 10 individuals before and after a short vacation. The stress scores (on a scale) are:

Subject	1	2	3	4	5	6	7	8	9	10
Before Vacation	48	51	47	53	49	56	50	45	52	50
After Vacation	46	50	48	52	48	57	49	47	51	51

We want to test whether the vacation changes the mean stress score. Use a two-tailed paired t-test at $\alpha = 0.05$.

Paired Two-sample T-test Example Solution

Hypotheses

$$H_0 : \mu_d = 0 \quad \text{vs.} \quad H_a : \mu_d \neq 0$$

where $d = (\text{After} - \text{Before})$ for each individual.

Step 1: Compute the differences. Compute $d_i = \text{After}_i - \text{Before}_i$ for each subject:

i	1	2	3	4	5	6	7	8	9	10
d_i	-2	-1	+1	-1	-1	+1	-1	+2	-1	+1

Step 2: Summary statistics of the differences.

$$\bar{d} = -0.3$$

The standard deviation $s_d \approx 1.479$.

Step 3: Standard error of the mean difference.

$$\text{SE} = \frac{s_d}{\sqrt{n}} = \frac{1.479}{\sqrt{10}} \approx 0.468$$

Paired Two-sample T-test Example Solution

Step 4: Compute the t statistic.

$$t = \frac{\bar{d} - 0}{SE} = \frac{-0.3}{0.468} \approx -0.641$$

Step 5: Degrees of freedom.

$$df = n - 1 = 10 - 1 = 9$$

Paired Two-sample T-test Example Solution

Step 6: Decision rule / p-value.

- The two-tailed critical t-value at $\alpha = 0.05$ is about ± 2.262 .
- Our test statistic $t = -0.641$ lies inside $[-2.262, +2.262]$. Thus, we **fail to reject** the null hypothesis H_0 .

Step 7: Conclusion.

There is no statistically significant evidence at the 5% level to conclude a change in mean stress score before vs. after the vacation. The sample mean difference of -0.3 could reasonably be due to chance.

5. Fisher's Exact Test (2×2 Contingency)

Concept:

- Used to determine if there is a non-random association between two categorical variables in a 2×2 contingency table.
- It is an **exact test**, meaning it computes the exact probability (P-value) of observing the data, assuming the row and column totals are fixed.
- **Bioinformatics Use:** Gene enrichment analysis. E.g., is a set of differentially expressed genes significantly **associated** with a specific biological pathway?

Contingency Table Structure:

	DE Genes	Non-DE Genes	Row Total
In Pathway	a	b	$a + b$
Not In Pathway	c	d	$c + d$
Column Total	$a + c$	$b + d$	$n = a + b + c + d$

Fisher's Exact Test Formula (Hypergeometric)

The probability of observing any specific table, given the fixed marginal totals, is calculated using the Hypergeometric Distribution:

$$P(a) = \frac{\binom{a+b}{a} \binom{c+d}{c}}{\binom{n}{a+c}} = \frac{(a+b)!(c+d)!(a+c)!(b+d)!}{a! b! c! d! n!}$$

P-value Calculation:

- The P-value is the sum of the probabilities of all tables that are **as extreme or more extreme** in the direction of the alternative hypothesis (H_1).
- **Example:** To test for positive association (H_1 : genes are enriched), we sum $P(a_{\text{observed}}) + P(a_{\text{observed}} + 1) + \dots + P(a_{\text{max}})$.

Fisher's Exact Test Example (Gene Enrichment)

Scenario: Test if a set of 120 differentially expressed (DE) genes is enriched in a pathway that contains 180 genes. (Total genes tested: 10,000).

	DE Genes	Non-DE Genes	Total
In Pathway	10	170	180
Not In Pathway	110	9710	9820
Column Total	120	9880	10000

Goal: Calculate the probability of observing 10 or more DE genes in the pathway by chance.

The Problem of Multiple Testing (The Multiple Hypothesis Trap)

- In Bioinformatics, we often test thousands of hypotheses simultaneously (e.g., $m = 20,000$ genes).
- If we use a significance level of $\alpha = 0.05$ for each test independently:
 - We expect $20,000 \times 0.05 = 1,000$ tests to be significant **by random chance alone**.
- This is an unacceptably high rate of **False Positives** (Type I Errors).
- **Solution:** Use an **Adjusted P-value** to control the overall error rate across the entire experiment (or 'family' of tests).

Error Rate Definitions

1. Family-Wise Error Rate (FWER):

- The probability of making **at least one** Type I error (false rejection) in the family of tests.
- **Goal:** Control $\text{FWER} \leq \alpha$.
- **Method: Bonferroni Correction.** (Very strict/conservative).

2. False Discovery Rate (FDR):

- The expected **proportion** of false discoveries (rejected H_0 's that are actually true) among all discoveries (all rejected H_0 's).

$$\text{FDR} = E \left[\frac{\text{False Discoveries}}{\text{Total Discoveries}} \right]$$

- **Goal:** Control $\text{FDR} \leq \alpha$.
- **Method: Benjamini-Hochberg (BH) Procedure.** (More powerful/less conservative).

Bonferroni Correction (Controlling FWER)

Method: Adjusts the significance threshold for each individual test.

- **Adjusted P-value Formula:**

$$p_{\text{adj}} = \min(m \times p_{\text{raw}}, 1)$$

where m is the total number of tests.

- **Interpretation:** You compare the adjusted P-value (p_{adj}) to the original α (e.g., 0.05).

Example: $m = 10,000$ genes, $\alpha = 0.05$.

- **Raw $p = 1 \times 10^{-5}$ (0.00001):** $p_{\text{adj}} = 10,000 \times 0.00001 = 0.1$.
- Since $0.1 > 0.05$, the result is **not significant** after Bonferroni correction.

Benjamini-Hochberg (BH) Procedure (Controlling FDR)

Method: A step-up procedure used for controlling the FDR.

- 1 **Sort:** Sort the raw p-values (p_i) in **ascending order**:

$$p_{(1)} \leq p_{(2)} \leq \cdots \leq p_{(m)}.$$

- 2 **Find k:** Find the largest rank k such that the raw p-value $p_{(k)}$ is less than or equal to its critical value:

$$p_{(k)} \leq \frac{k}{m}\alpha$$

- 3 **Reject:** Reject all null hypotheses from rank 1 up to k .

Adjusted P-value (q-value) Formula: The q -value for the i -th ranked p-value is calculated as:

$$q_{(i)} = \min_{j \geq i} \left(p_{(j)} \times \frac{m}{j} \right)$$

Advantage: BH is more powerful than Bonferroni, making it the preferred method in large-scale biological experiments (e.g.,

- Maryam Tavasoli
- Sahand Akramipour

References

- <https://www.geeksforgeeks.org/data-science/z-test/>
- <https://www.geeksforgeeks.org/data-science/t-test/>
- <https://www.geeksforgeeks.org/maths/fisher-exact-test/>
- <https://www.jmp.com/en/statistics-knowledge-portal/t-test/one-sample-t-test>
- <https://www.statology.org/paired-samples-t-test>
- <https://en.wikipedia.org/wiki/Fisher>
- Additional references for Permutation Test, Bonferroni, and Benjamini-Hochberg.

The End

Questions? Comments?